04T: Prediction for SLR

Stat 230 - Spring 2026

> **Note**
>
> This is the tutorial version of the R activity. If you prefer to use .qmd and RStudio directly, you can use https://aluby.github.io/stat230/activities/04R-prediction

1 Loading data

The data set is loaded by running the following code chunk:

```
city <- readr::read_csv(url("https://raw.githubusercontent.com/zief0002/modeling/mair
```

The columns of interest are `education` and `income`.

2 Making plots

Q1. Make a plot of `income` against `education` and overlay the linear regression equation. Does it look like what you expected? (I've started the code for you; you will need to overlay the linear regression equation)

```
#| min-lines: 3
gf_point(income ~ education, data = city)
```

Q2. Make a plot of `income` against `scale(education)` and overlay the linear regression equation. Does it look like what you expected?

```
#| min-lines: 3
# Write your graph code here
```

3 Fitting a simple linear regression model

Q3. Use the `lm()` command to fit the simple linear regression model where `education` is used to predict `income`. Make sure your output matches the output on the worksheet.

```
#| min-lines: 3
# Write your lm() command here
```

4 Making predictions

The first person in this dataset has `education = 8`.

Q4. Calculate the expected income of this person using the first regression equation. (You may use R or do this by hand. There's a code chunk below for your convenience)

```
#| min-lines: 3
# Write your code here
```

Q5. If we want to predict the income of **this** person, should we use a confidence interval or a prediction interval?

Q6. Use R to construct the appropriate 89% interval for this person's income.

```
#| min-lines: 3
# Write your code here
```

Q7. Interpret the interval in context.

5 Plotting our intervals

It can be useful to plot your confidence/prediction intervals to communicate your findings to a broader audience. The code below creates a scatterplot of the data, adds the fitted regression line, and then adds 89% prediction intervals.

```
gf_point(income ~ education, data = city) |>
  gf_lm(interval = "prediction", level = .89) |>
  gf_lm(interval = "confidence", alpha = 0.6, level = .89)
```

Q8. Run the code to produce a scatterplot, regression line, and both types of intervals. Which is the prediction interval and which is the confidence interval? How can you tell?

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6 Function quick reference

The following table summarizes the functions we learned today:

Function	Purpose
`lm(formula, data)`	Fit a linear model
`predict(model, newdata, interval, level, se.fit)`	Make predictions and calculate intervals
`gf_point(formula, data) |> gf_lm(interval = "confidence")`	Plot data and add confidence intervals for the mean response
`gf_point(formula, data) |> gf_lm(interval = "prediction")`	Plot data and add prediction intervals for the mean response